



2026 高科技廠房設施國際論壇 High-Tech Facility International Forum 2026

打造 AI-Ready Data Center : OT 通訊與資安如何重塑廠房營運韌性

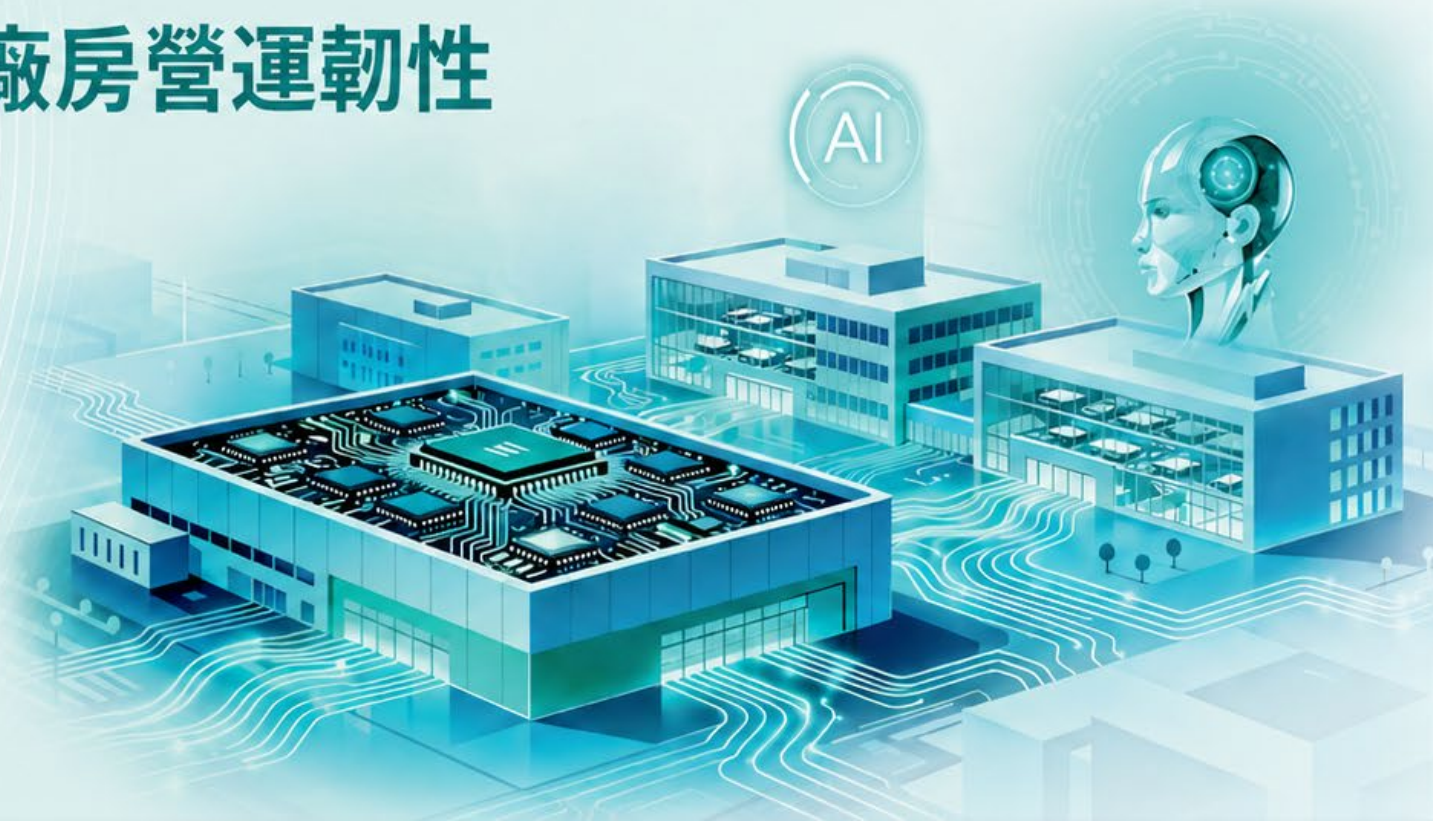
Building an AI-Ready Data Center:
How OT Networking and Security
Reshape Operational Resilience

講者 Speaker

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Cisco PM

Thursday, September 3, 2026



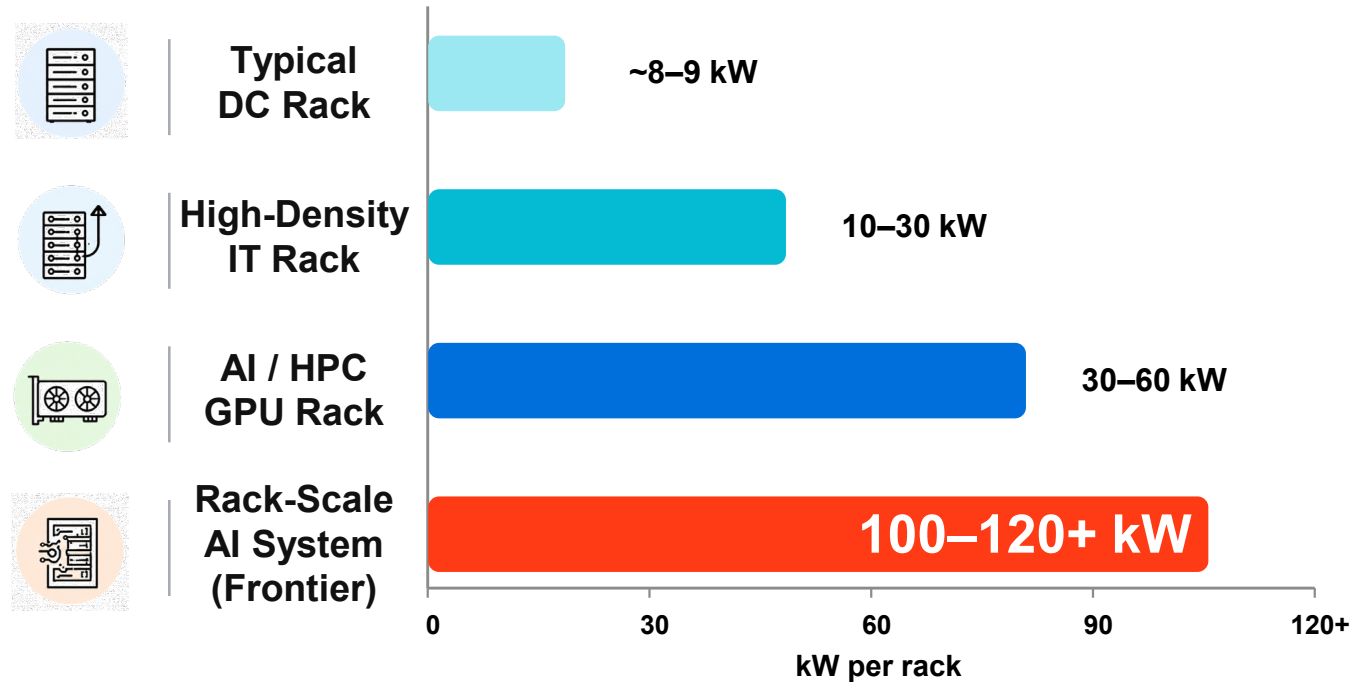
The Growing Complexity of AI Data Centers Driven by Compute and Power Demands

AI Usage
Data Centers
Buildings
Energy Generation
Climate Change



AI Data Center Power: From Density to Operational Complexity

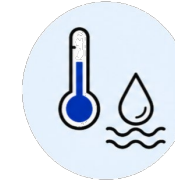
Rack Power Density Is Rising Fast



From ~8–9 kW typical racks to 100–120+ kW rack-scale AI systems, power and cooling become core design constraints—and real-time visibility and secure control become essential to operational resilience

Key Operational Challenges

1. Cooling Becomes Mission-Critical



Higher rack density drives liquid cooling, CDU, leak detection, and real-time thermal monitoring requirements.

2. Power Architecture Gets More Complex



BESS, EMS, renewables, UPS, PDU, and microgrids must be coordinated as one integrated system.

3. OT Visibility & Operational Resilience Become Essential



Real-time visibility and secure control are critical to preventing cascading failures and downtime.

Cybersecurity is now the #1 obstacle to AI adoption

AI-driven connectivity increases cybersecurity risks

2024

#1 | 42%

Shortage of skilled workers

#2 | 40%

Inflation

#3 | 37%

Cybersecurity risks

Q. What are the biggest external obstacles to your organization's growth?

2026

#1 | 40%

Cybersecurity concerns

#2 | 36%

Technology integration challenges

#3 | 34%

Lack of skilled talent

Q. What are the biggest obstacles your organization faces in adopting AI?

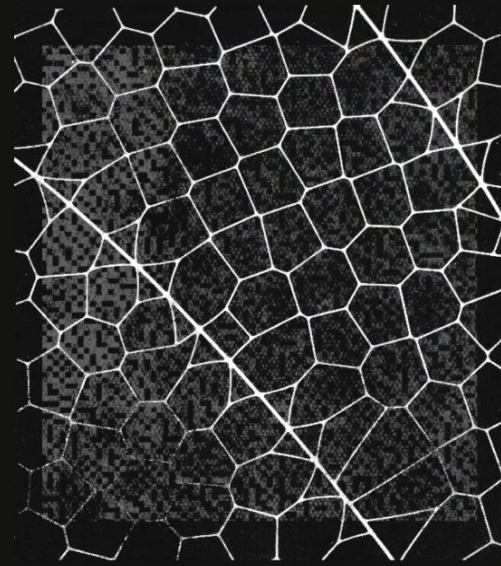


In 2024, cybersecurity emerged as a growing priority for industrial networking teams; today, it has become the leading concern, with 48% of organizations citing security as their top networking challenge.



AI Security: A Shared Industry Imperative

ANTHROPIC Project Glasswing



“There is an urgent need for as many organizations as possible to shore up their cyber defenses.”*



ANTHROPIC



BROADCOM



CROWDSTRIKE

Google

JPMorganChase



Microsoft

NVIDIA

paloalto
NETWORKS

Project Glasswing Cyber Defenders

*Project Glasswing: An initial update, May 22, 2026 - Anthropic



Cyber Risk in the AI Data Center

Attacks can come from **outside** — or **start inside** and **move laterally**.

1 EXTERNAL THREATS (OUTSIDE-IN)

- Unauthorized Access
- Internet Exploits



INTERNET



STANDARD SWITCH



WI-FI AP



CONNECTED DEVICES



IP CAMERA

2 INSIDER / AI DEVELOPER RISK (INSIDE-OUT)

AI-generated or misguided code introduced into the Air Handling System.



KEY SECURITY GAPS



NO VISIBILITY

- Blind to assets, vulnerabilities
- Cannot detect abnormal activities



NO SEGMENTATION

- Flat network between IT and OT
- High risk of lateral movement



NO THREAT DETECTION

- No monitoring for threats
- Intrusions go undetected until damage occurs

POTENTIAL IMPACT



COOLING FAILURE
Overheating and downtime



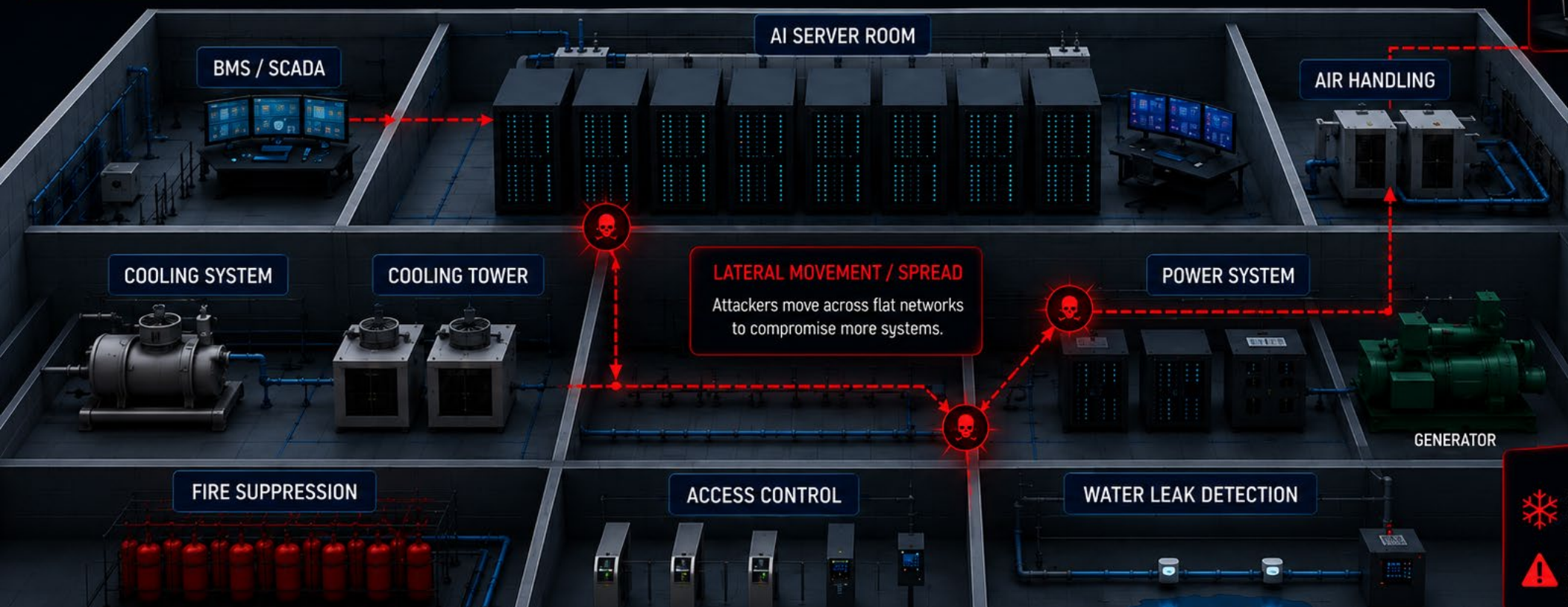
POWER DISRUPTION
Unexpected downtime and system damage



SERVICE OUTAGE
AI workloads interrupted
Revenue and SLA impact



BUSINESS LOSS
Reputation damage and financial loss



Next-Generation OT Infrastructure Strategy for the AI Era

▪ Globally Validated Architecture

AI data centers are moving toward standardized, repeatable, and scalable architectures. IT, OT, energy, cooling, and cybersecurity systems must work under one framework to support efficiency, security, and operational resilience in high-density computing environments.

▪ Industrial Networking for AI Operations

AI applications require real-time collaboration across OT controllers, energy systems, cooling equipment, and analytics platforms. Future data center networks must go beyond connectivity, providing predictability, high availability, secure segmentation, and continuous power support.

▪ High Availability with Network Redundancy

AI infrastructure has very low tolerance for downtime, requiring uninterrupted connectivity and rapid recovery. PRP / HSR, TSN, and industrial wireless enable stable, low-latency communication for robots, AGVs, and mobile equipment in high-interference environments.

▪ Visibility and Compliance Readiness

AI computing increases energy and cooling loads, making real-time visibility into equipment status, power usage, and abnormal events essential. Visibility, anomaly detection, and secure remote access help reduce troubleshooting time and alignment with standards such as IEC 61850 and IEC 62443.

AI Data Center Architecture

- **Proven Industry Architecture**

Based on global best practices across manufacturing, energy, and critical infrastructure.

- **High Availability Operations**

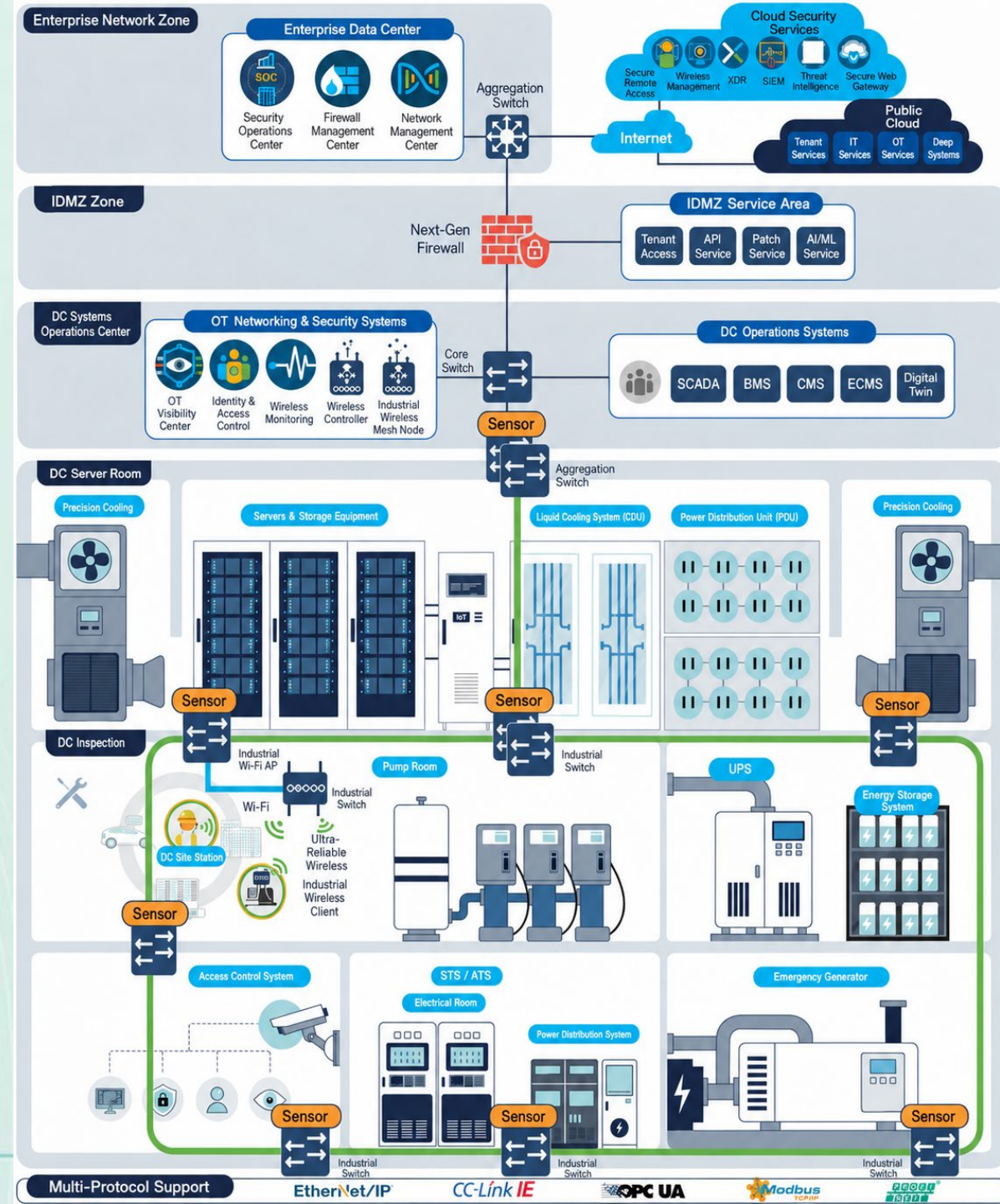
Supports REP, PRP, MRP, and HSR for uninterrupted operations.

- **Open OT Integration**

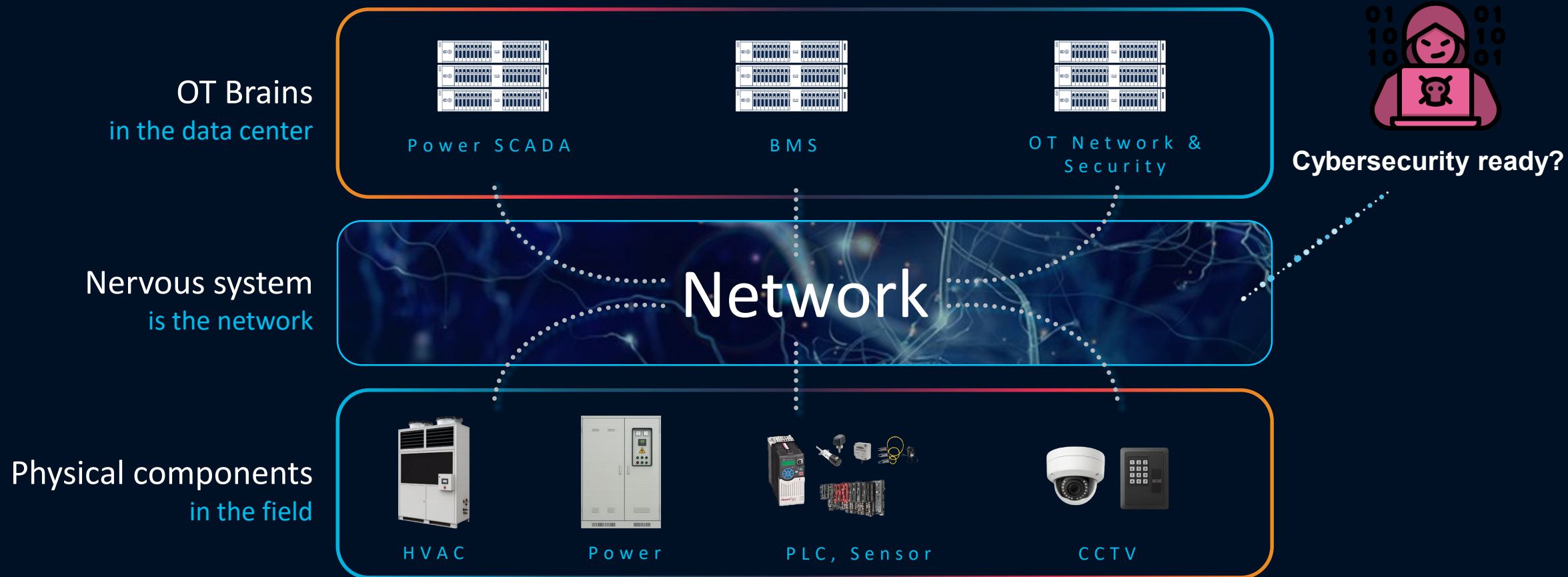
Supports IEC 61850, PROFINET, EtherNet/IP, and OPC UA.

- **IEC 62443 Security Framework**

Asset visibility, network segmentation, and Zero Trust remote access.



The First Step: Re-Architect AI Data Centers with a Three-Layer Model



Network + Visibility: The Foundation of OT Cybersecurity

- **Connected Asset Visibility**

Automatically discovers and visualizes all OT/IT devices, providing a clear view of exactly what exists in the factory.

- **Identify and Track Vulnerabilities**

Analyzes firmware versions, protocols, and vendors, and continuously tracks risks against the CVE vulnerability database.

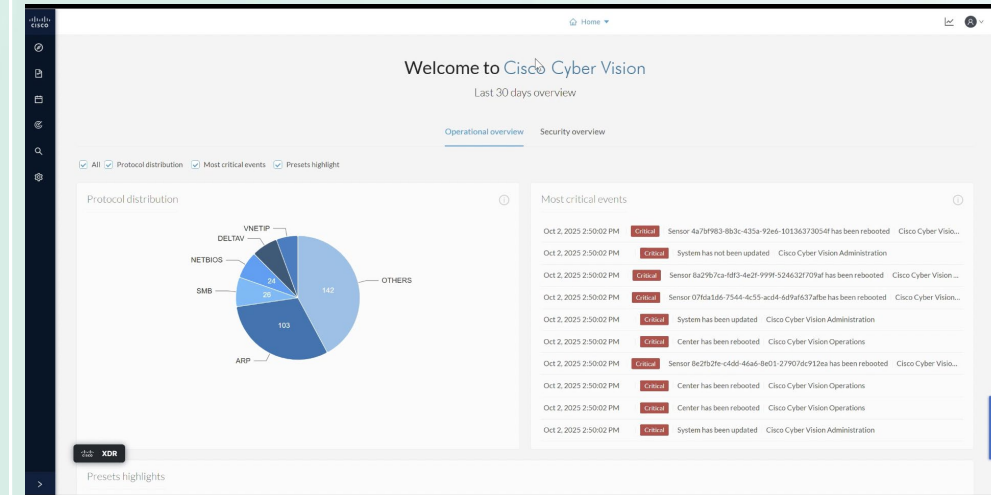
- **Track Risk Exposure**

Provides risk assessments so managers can identify which assets are the most vulnerable and most likely to be targeted.

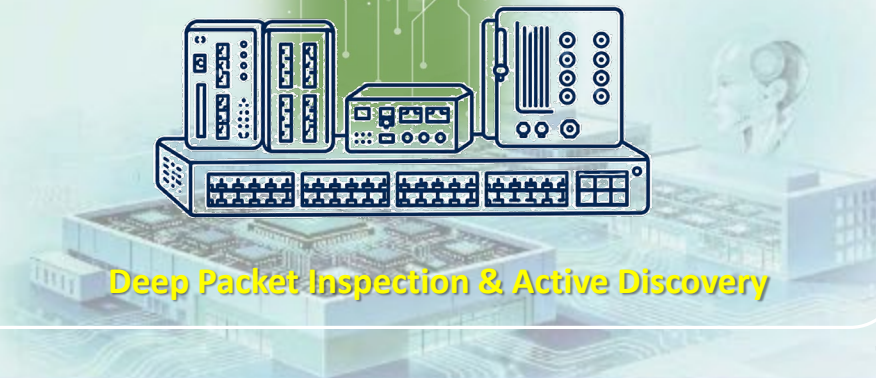
- **Detect Malicious Intrusions**

Uses DPI and threat intelligence to detect abnormal or malicious activity at an early stage.

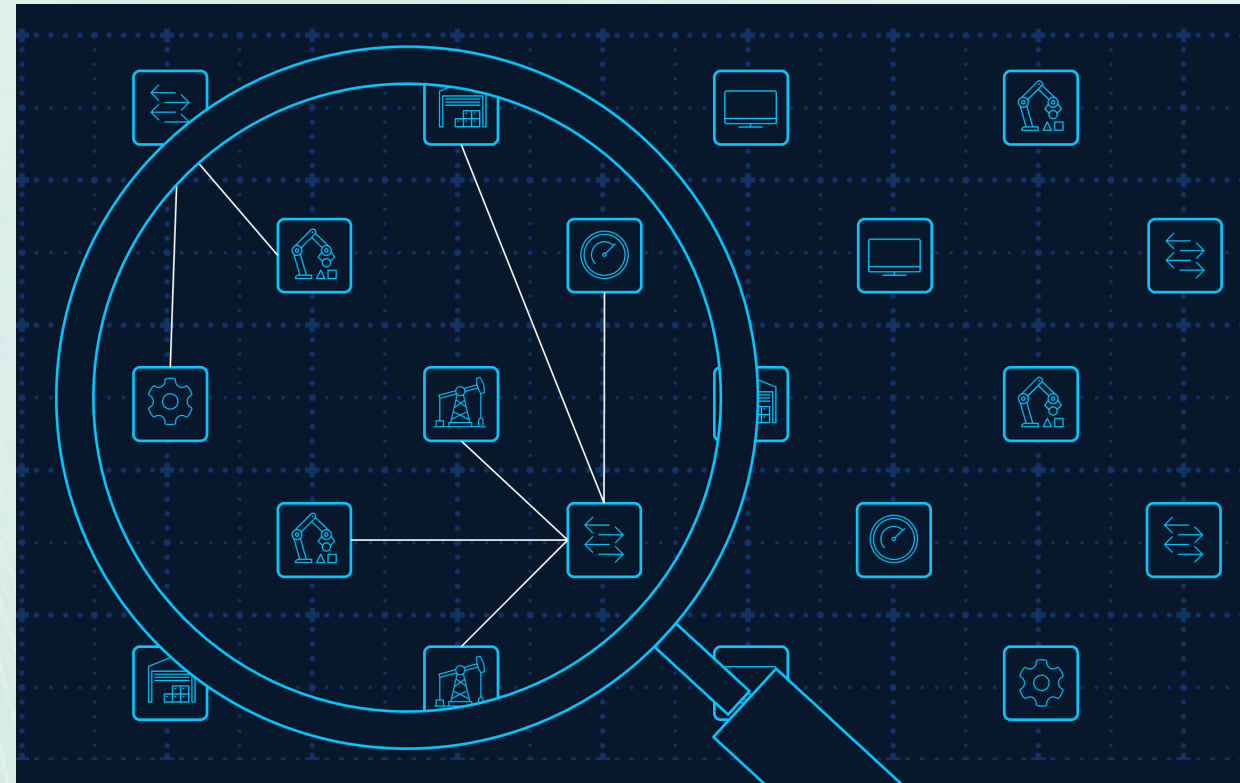
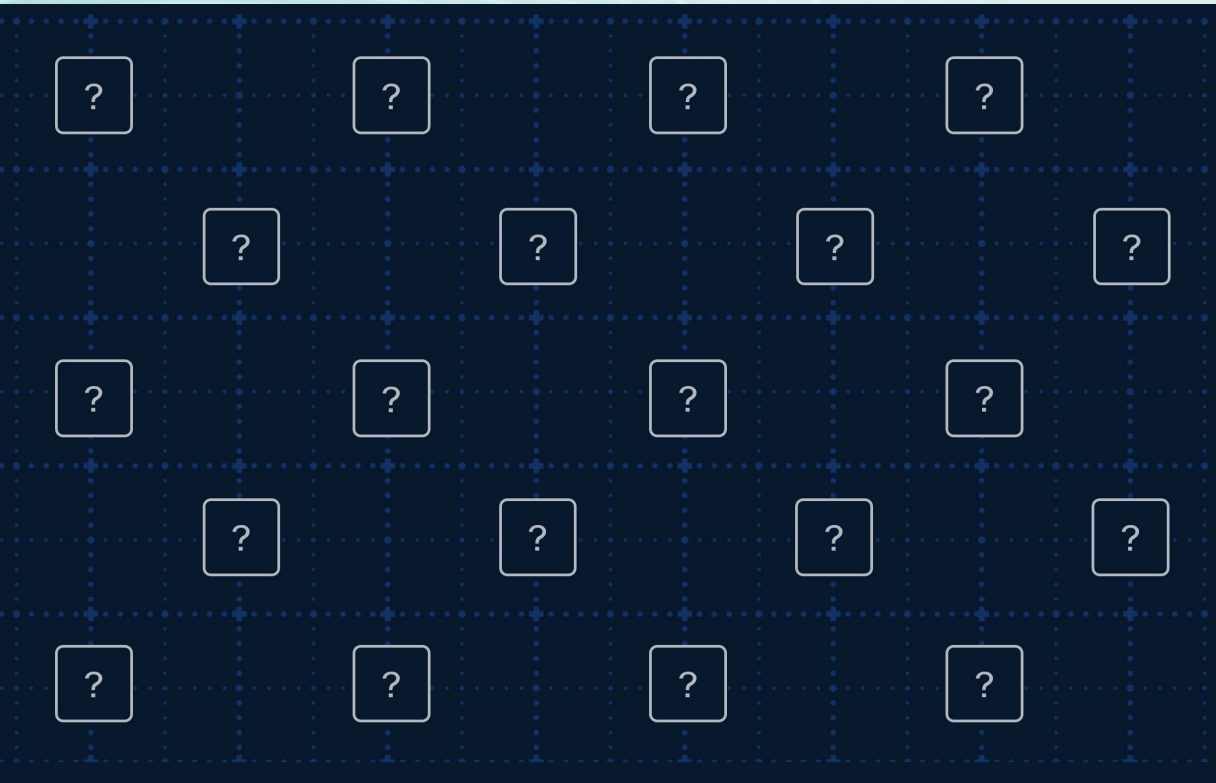
OT Visibility



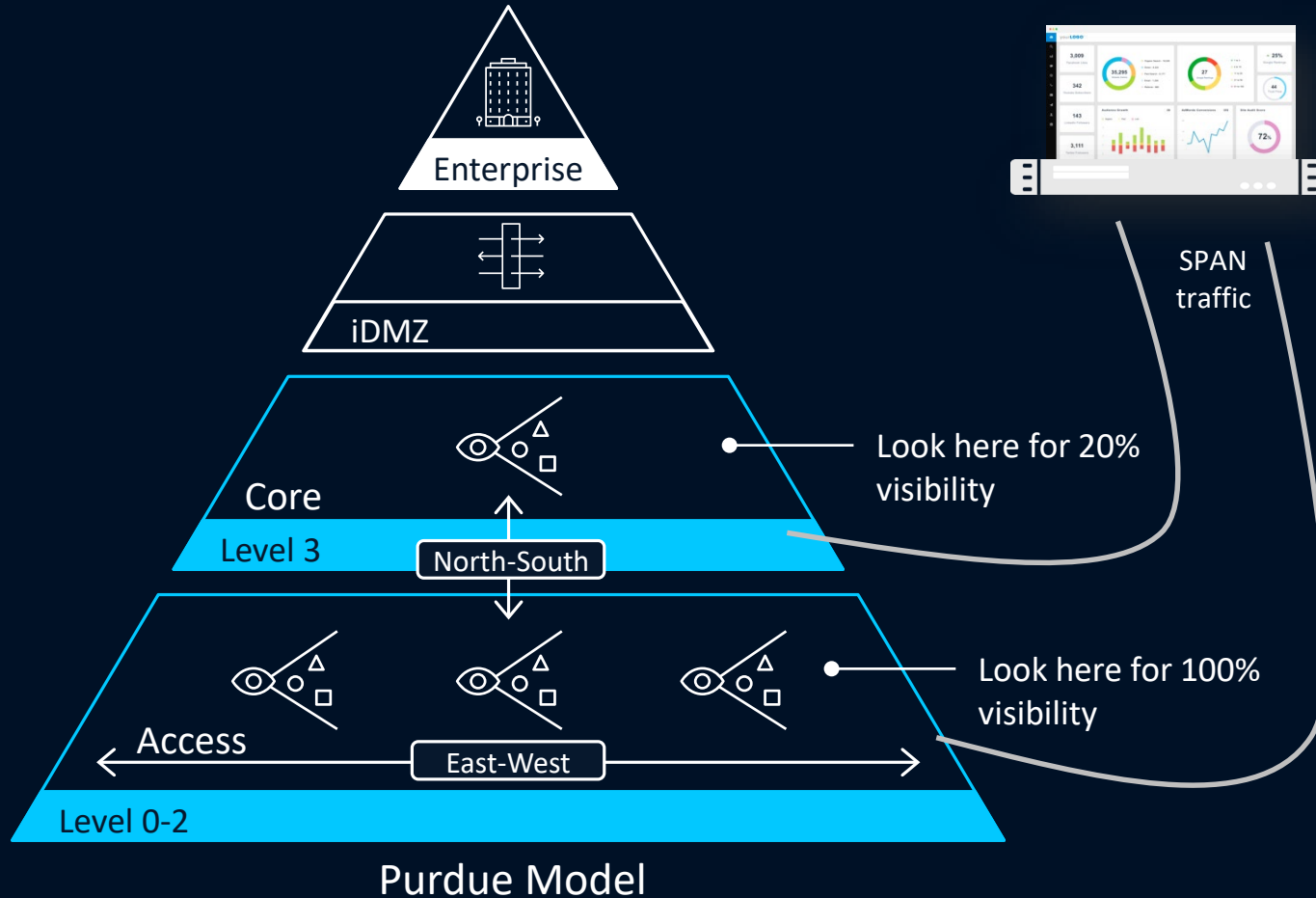
AI Analysis



From Unknown to Visible



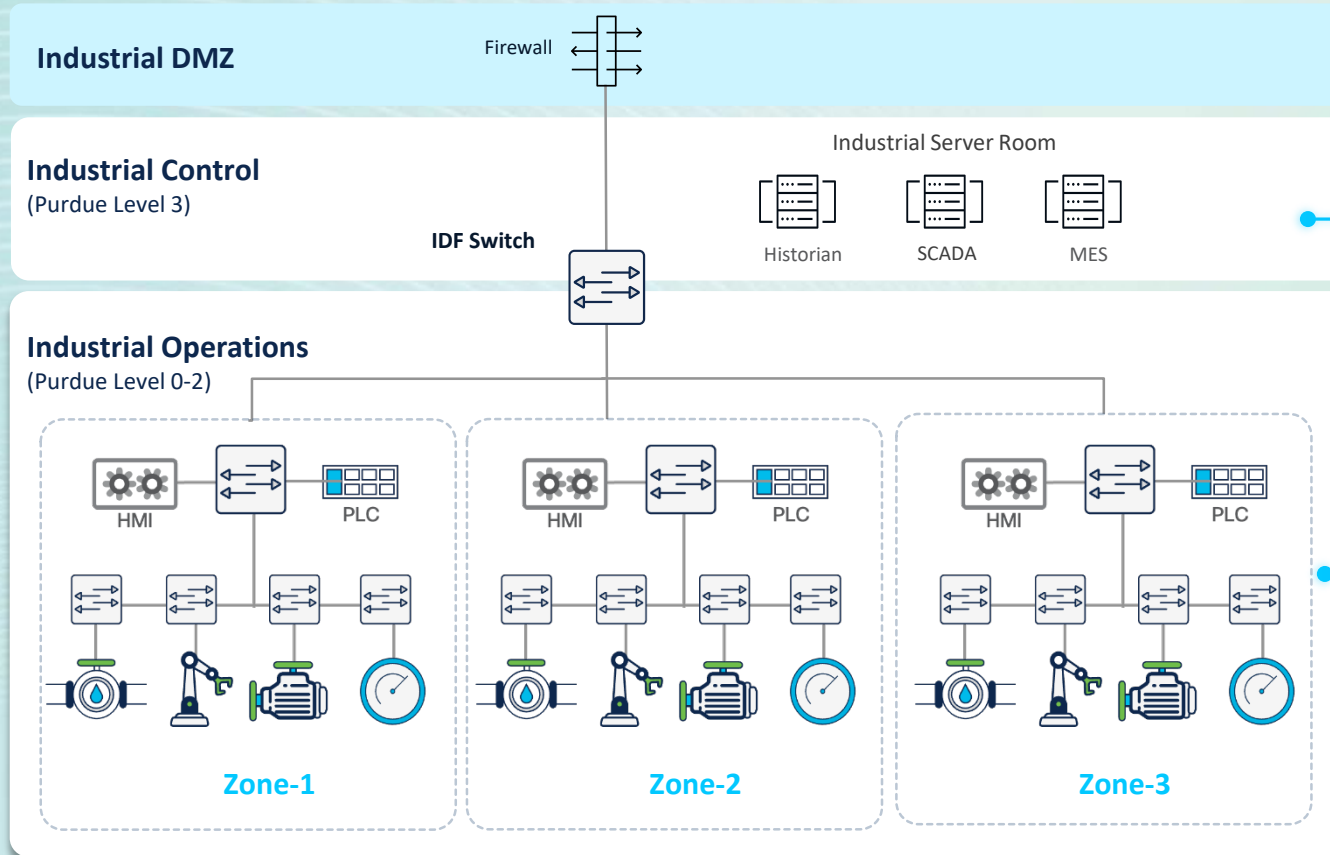
Practical Lesson: OT Visibility Depends on Where You Inspect Traffic



Core-level monitoring provides only North-South visibility, leaving East-West OT traffic unseen.

Full OT visibility via SPAN or dedicated sensors is expensive and operationally complex.

From OT Visibility to Zero Trust: Making Micro-Segmentation Practical



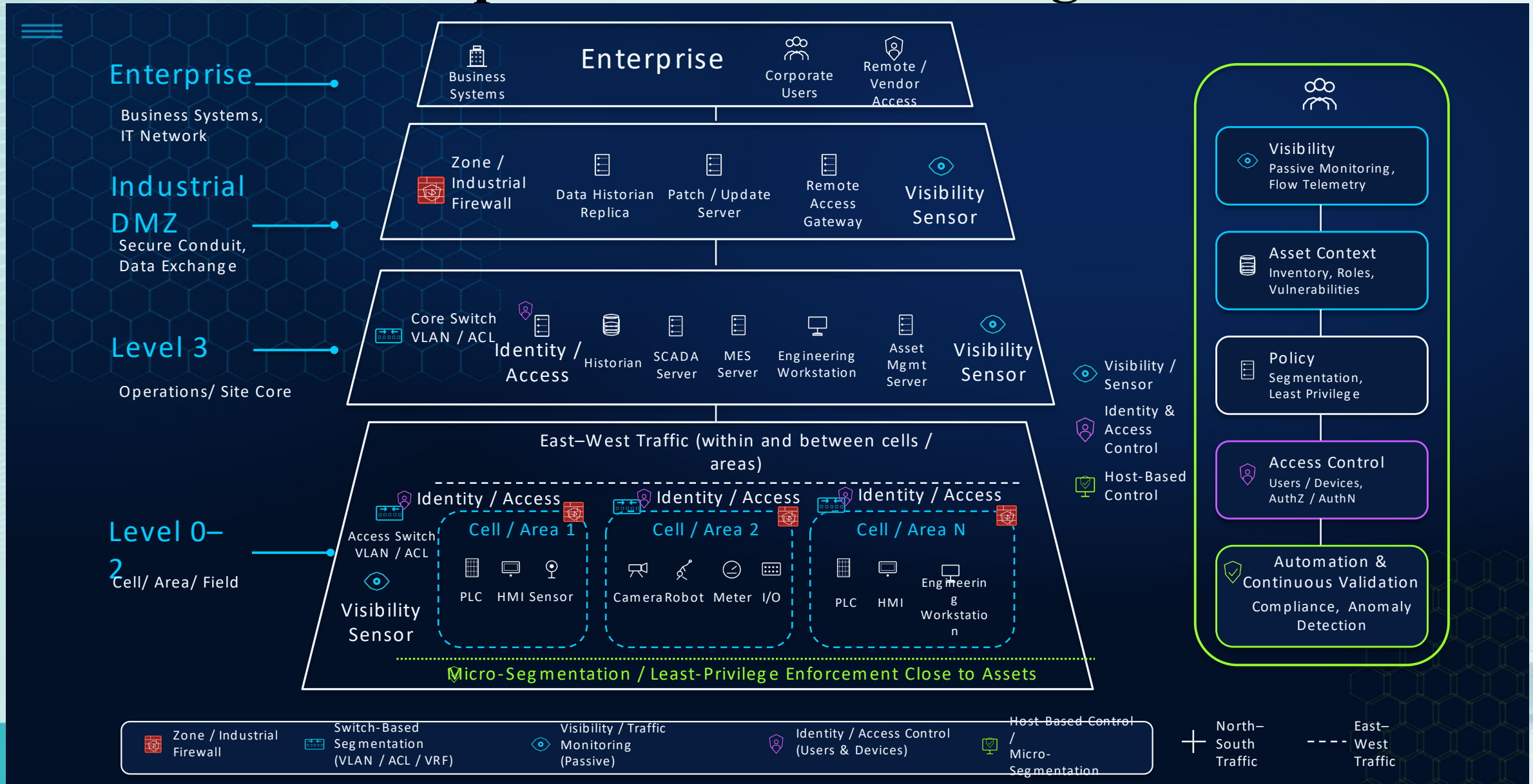
Blind below the IDF

Gaining visibility at Levels 0-2 using SPAN or packet forwarders is expensive

Fear of unplanned downtime

Without visibility, you run the risk of blocking legitimate process flows

Paths to Implement OT Micro-Segmentation



Paths to Implement OT Micro-Segmentation



This user interface understands industrial processes. I can group assets into zones



I now have OT context to build the right security policies

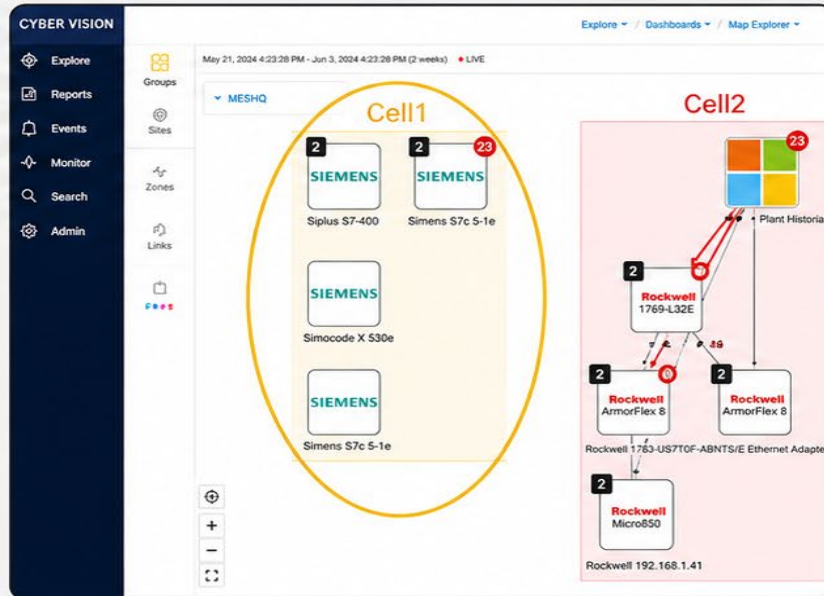
Operational Impact

Target KPIs to be validated during Project

80%+
Asset visibility coverage

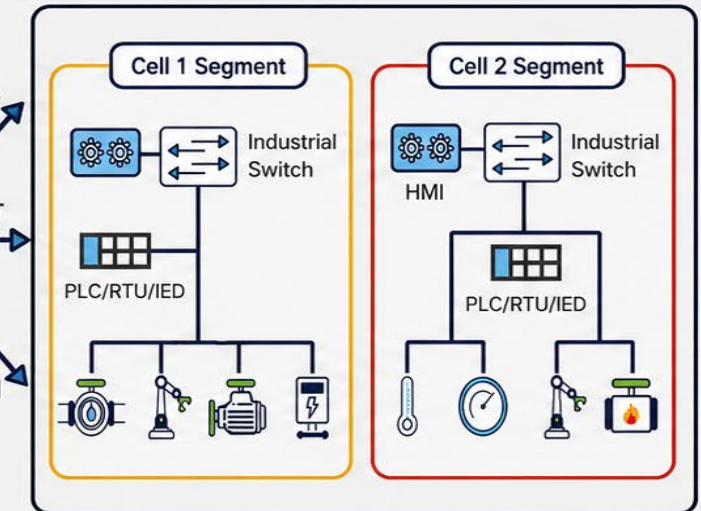
90%+
Unauthorized east-west traffic detected

70–90% ↓
Segmentation policy deployment time



	Cell 1	Cell 2	PLC	MES
Cell 1	✓	✗	✓	✗
Cell 2	✗	✓	✓	✗
PLC	✓	✓	✓	✓
MES	✗	✗	✓	✓

Asset identity and zone attributes are shared with the policy engine for automated segmentation.



A Practical Journey to OT Cyber Resilience

Prevent & Protect

Detect & Respond



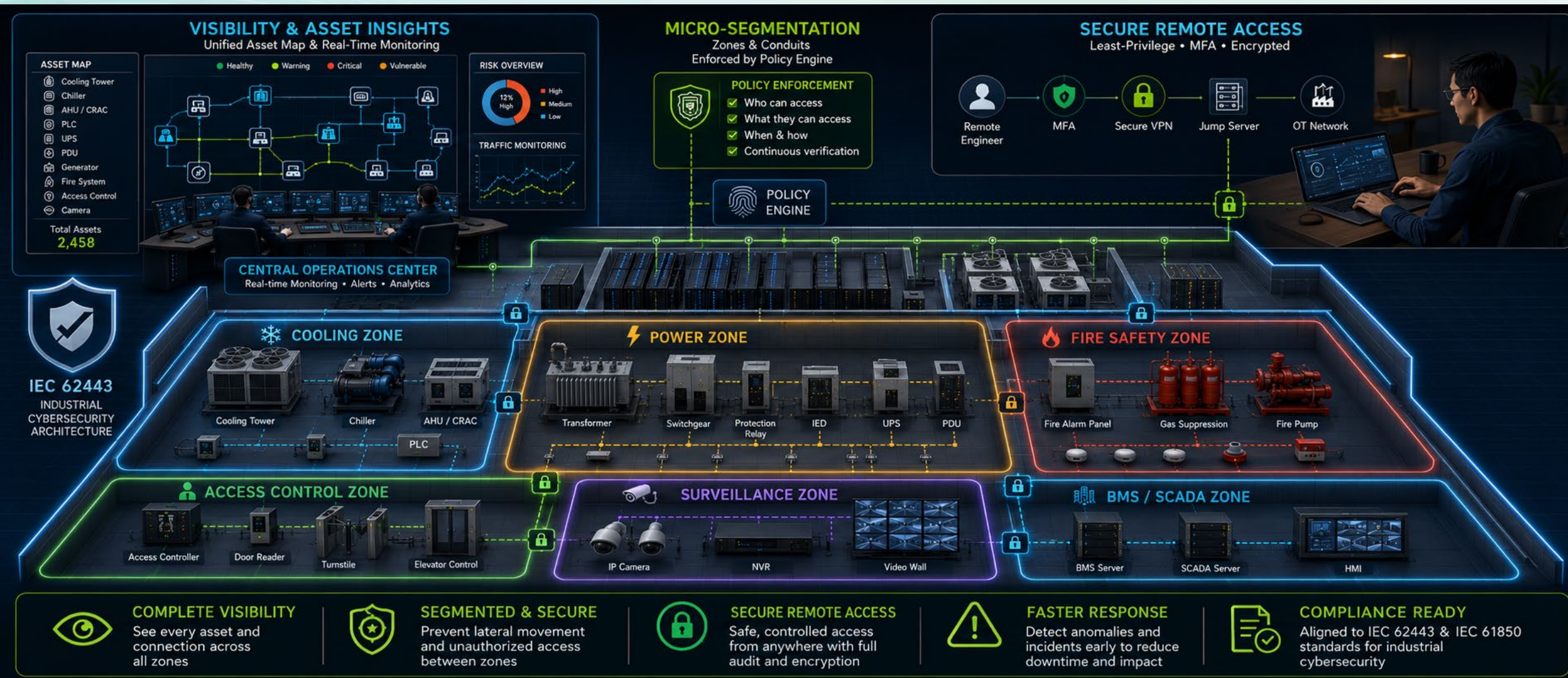
Visibility to your risk exposure

Segmentation to limit the blast radius

Secure remote access to OT assets

SOC monitoring for your OT network

Begin Your Journey to OT Resilience



Q&A

